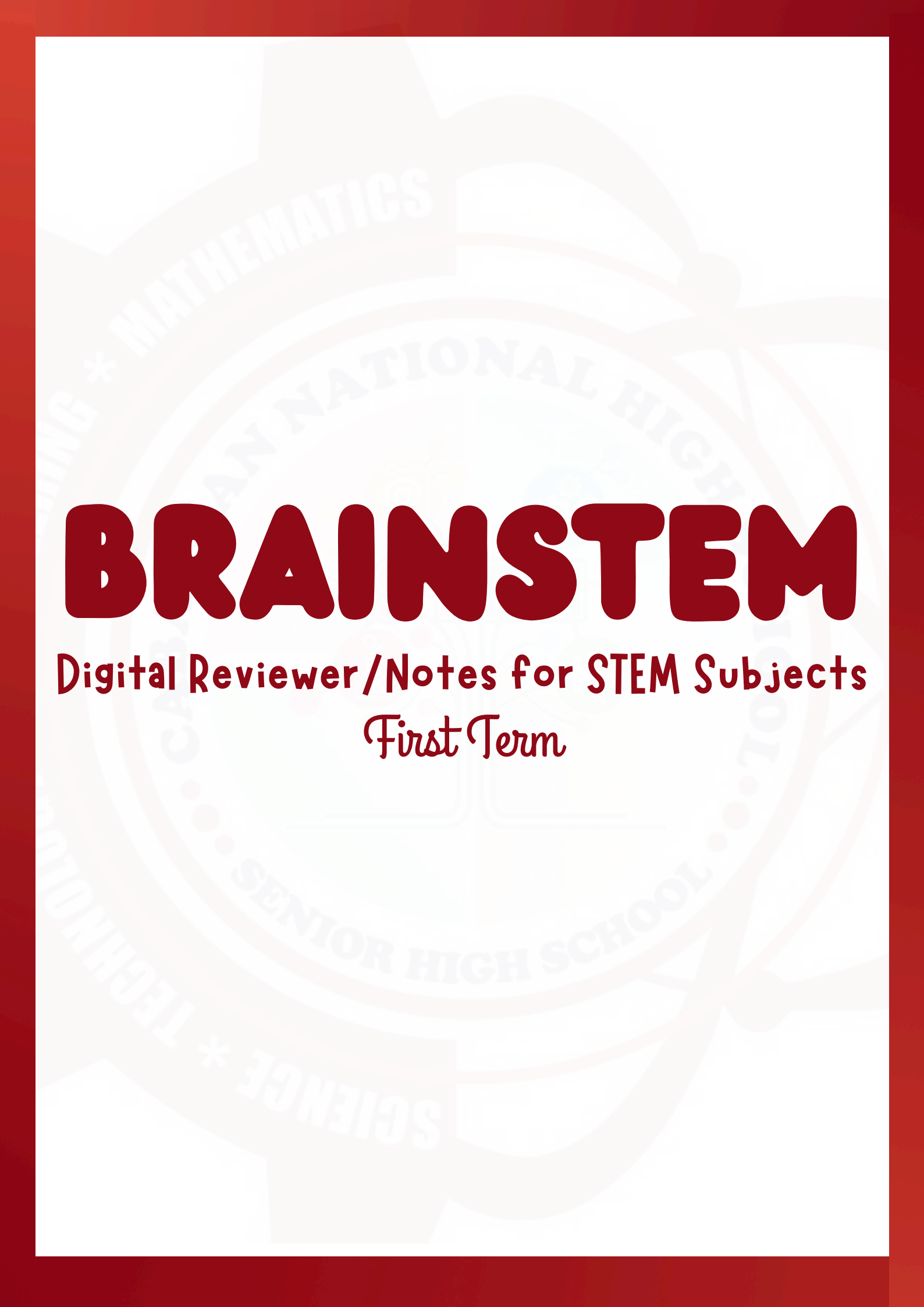




BRAINSTEM

Digital Reviewer/Notes for STEM Subjects
First Term



Quantitative Research

–uses scientifically collected and statistically analyzed data to investigate observable phenomena

Characteristics:

- **Large Sample Size**– to obtain more meaningful statistical result, the data must be come from a large sample size
- **Objectivity**– data gathering and analysis of results are done accurately, objectively, and are unaffected by the researcher's intuition and personal guesses.
- **Concise Visual Presentation**– data is numerical which makes presentation through graphs, charts, and tables possible and with better conveyance and interpretation
- **Faster Data Analysis**– the use of a statistical tools gives way for less time-consuming data analysis
- **Generalized Data**– data taken from a sample can be applied to the population if sampling is done accordingly, i.e., sufficient size and random samples were taken
- **Fast and Easy Data Collection**– quantitative research uses standardized research instruments that allow the researcher to collect data from a large sample size efficiently
- **Reliable Data**– data is taken and analyzed objectively from a sample as a representative of the population, making it more credible and reliable for policymaking and decision making
- **High Replicability**– can be repeated to verify findings enhancing its validity, free from false or immature conclusions.

Advantages of Quantitative Research:

- Very objective
- Numerical and quantifiable data can be used to predict outcomes
- Findings are generalizeable to the population
- There is conclusive establishment of cause and effect
- Fast and easy data analysis using statistical software
- Fast and easy data gathering
- Can be replicated or repeated
- Validity and reliability can be established

Disadvantages of Quantitative Research:

- Lack necessary data to explore a problem or concept in depth
- Does not provide comprehensive explanation of human experiences
- Some information cannot be described by numerical data such as feelings, and beliefs
- The research design is rigid and not very flexible

- The participants are limited to choose only from the given responses
- The respondents may tend to provide inaccurate responses
- A large sample size makes data collection more costly

Kinds of Quantitative Research

1. Descriptive Design

- use to **describe** a particular phenomenon by observing it as it occurs in nature
- there is **no experimental manipulation**
- the researcher **does not start with a hypothesis**
- the goal is to describe the person or object of the study
- ex. "the determination of the different kinds of physical activities and how often high school students do it a monthly basis"

2. Correlational Design

- identifies the **relationship between variables**
- data is collected by observation since it does not consider the cause and effect
- ex. "the relationship between the amount of physical activity done and student academic achievement"

3. Ex post facto Design

- used to investigate a possible relationship between **previous events and present conditions**
- "ex post facto" means after the fact, looks at the possible causes of an already occurring phenomenon
- ex. "how does the parent's academic achievement affect the children obesity?"

4. Quasi-experimental Design

- used to **establish the cause-and-effect relationship** of variables
- has lesser validity than experimental design due to absence of random selection and assignment of subjects
- the independent variable is identified but **not manipulated**
- the researcher does not modify pre-existing groups of subjects
- the group exposed to treatment (experimental) is compared to the group unexposed to treatment (control)

4. Experimental Design

- like quasi-, it is used to establish the cause-and-effect relationship of two or more variables
- provides a **more conclusive result** because it uses random assignment of subjects and experimental manipulations.



Importance of Quantitative Research Across Fields

Field	Contribution/Application	Example
SOCIAL SCIENCE	- show effects of intervention to the group behavior - understand cultural or racial conflicts - human satisfaction	the effects of pandemic and social behavior and economic stability
NATURAL AND PHYSICAL SCIENCES	- investigate the effectiveness of a product or treatment to illnesses - finding or enhancing alternative energy sources - advancement in material science	antidiabetic properties of common Philippine herbs
AGRICULTURE AND FISHERIES	- increase yield of crops - prevent and cure crops and livestock diseases	the effectiveness of organic and inorganic fertilizer to vegetable production
SPORTS	enhance athletic performance	diet and exercise techniques for different kinds of sports
BUSINESS	- offer device marketing strategies - improve marketability	effectiveness of facebook ads on sales
ARTS AND DESIGN	- show relationship between color and architectural space - maximize use of multimedia and adaptation for recreation, business marketing and lifestyles changes	the effects of music on learning and behavior
ENVIRONMENTAL SCIENCE	determine cause and effects of climate change	the environmental factors affecting natural calamities

PRACTICAL RESEARCH 2



Designing Research topic

1. Chose a broad topic

- A. Chose an interesting topic
- B. Select a significant topic
- C. Choose a topic relevant to your field
- i. Resources
- ii. Review of Literature
- iii. Field experts
- iv. Brainstorming

2. Do a preliminary research

- reading some more articles, journals, and related research studies
- find out how other researchers gathered their data, what research instruments were used, how the data were analyzed, and what important findings they shared

3. Define a problem

- list some questions or problems that you want to research
- narrow down broad topic into feasible and manageable research questions

4. Refine the question

- evaluate the questions formulated

BASIC GUIDELINES IN MAKING A RESEARCH STUDY TITLE

1. Use an **accurate description** of the subject and scope of the study instead of using general terms.
2. **Do not use abbreviations** except for commonly known ones like DNA and ICT.
3. Do not include words like "The study of," "Analysis of," "An investigation of," or similar construction as these would only lengthen the title.
4. Include the main **dependent and independent variables**.
5. Be mindful of the **proper use of grammar and punctuations**.
6. **Capitalize** all nouns, pronouns, verbs, adjectives, adverbs, as well as first letter of the first and last words.
7. State in a **declarative form**.
8. The year the study has been conducted **should not be included** unless it is a historical study.
9. Use **current** terminology.
10. Depending on the institutional requirements, **5 to 15 words** are sufficient to describe the research study.
11. Use the **common name** instead of chemical formula.
12. **Write and italicize** full scientific name.
13. Make sure to **reflect the tone** of the paper.

STEPS IN WRITING YOUR RESEARCH TITLE

1. Determine **what it is that you wish to accomplish** or know from your study. Write one to two sentences to state the main objectives of your research projects.
2. Include important **keywords and variables**. Revise the sentences into one complete sentence that include important keywords and variables of the study.

3. **Shorten the title** by eliminating unnecessary words. You may also shrink a phrase into a simpler phrase or a single word. In doing this, make sure that the main thought of the research study is retained.
4. **Correct grammar and punctuation** errors if there is any.
5. Observe **proper formatting**. The format may vary according to the requirements of the course or school.

BACKGROUND OF RESEARCH

- inform the reader of the **context** of the study
- **justify the need to conduct** a research study about the topic selected by establishing the research gap

Research Gap

- under or unexplored area of a topic that requires further exploration

Background of the Study

- purpose of **relaying the importance** of your research study

Review of Related Literature

- **more comprehensive** and thoroughly discusses the studies mentioned

Questions answered:

1. What is already known about the topic?
2. What is not known about the topic?
3. Why do you need to address those gaps?
4. What is the rationale of your study?

RESEARCH QUESTIONS

Statement of the Problem

- introduces the **problem** that you need to investigate or address

Research Question

- help **clarify and specify** the research problem
- also cons as **sub problems** of your research problem
- specify the method when collecting and analyzing data
- considered as the **main focus**

Characteristics of Good Research Questions Fraenkel and Wallen (2020)

1. FEASIBLE

- consider the amount of time, energy, money, respondents, and even your current situation as student-researcher

2. CLEAR

- clarity of how the questions are related lead to agreement of meaning of the readers of your study

3. SIGNIFICANT

- ask if your research question are relevant or important to ask

4. ETHICAL

- always consider the welfare of people, animals, or whoever are involved in your study

PRACTICAL RESEARCH 2



General Question

- derived from the research problem

General Question

- anchored in the general research problem

SCOPE AND DELIMITATION

Scope

- specifies the coverage of your study such as on variables, population or participant and timeline

Delimitation

- cite factors of your study that are not included or excluded or those you will not deal in your study.

CLEAR WRITTEN Scope and Delimitation

- make it easier to answer questions which are related or not related to your study

Components

Topic of the Study – What are the variables to be included and excluded?

Objectives or Problems to Addressed – Why are you doing this study?

Time Frame – When are you going to conduct?

Locale of the Study – Where are you going to conduct?

Characteristics of the Respondents – Who will be your respondents?

Method/Research Instruments – How are you going to collect the data?

Difference between Delimitation and Limitation

Delimitation – describe the various limitations that rose during the design and conduct of the study

Limitation – those variables included in your study; identified the scope of the study

DEFINITION OF TERMS

- helps simplify some of the technical terms that are vital in the understanding of the research
- advantageous to the reader to include
- prevent the ambiguous meaning of terms

Conceptual

- based on how it is defined in the dictionary or encyclopedia

Operational

- how the terms/word was used in the study

2 Cases in which important terms are defined

1. Not common or widely known
2. If the term has a specific or unique meaning in the context of the study

Benefits of Having Definition of Terms

1. It is a **useful place to include technical terms** in the topic of the research questions
2. It can **clarify the definition** of terms, especially if it has a different meaning. Define the term according to how it was used in the study.

3. It **makes it easier to revisit or check the meaning** of a term instead of trying to locate it through the paper.
4. Helps to **ensure that the reader can understand the technical terminologies** and jargons while reading the paper.

Guidelines on How to Write the Definition of Terms

1. Write a brief introducing statement. It must shortly **describe the content** of the definition terms.
2. List/write the words/terms (which are technical) that would be included (make sure that the variables and key terms found in the title are included).
3. The terms should be **arranged** to each term.
4. **Indention** should be applied to each term.
5. The term should be **followed by a period**.
6. It can be underlined or not.
7. It can be bold and italic or not.
8. It does not have to be lengthy (direct to the point).
9. Acronym/initials should be defined clearly.
10. Do not overflow with technical terms.
11. Keep the definition brief and basic.

RESEARCH FRAMEWORK

- illustrate the **structure or blueprint** of the research plan and helps the researcher formulate relevant Research Questions
- consist of **key concepts and assumes relationships** of the research project
- normally used as a **guide for researchers** so that they are more focused on the scope of their studies
- can be presented as visual or narrative form

THEORETICAL FRAMEWORK

- used for studies that **anchor on time-tested theories** that relate the findings of the investigation to the underpinning relevant theory of knowledge

CONCEPTUAL FRAMEWORK

- refers to the **actual ideas, beliefs, and tentative theories** that specifically support the study
- primarily a conception or model of what is out there that the researcher plans to study

Similarities of the Theoretical and Conceptual Framework

1. Provide an **overall view** of the research study
2. **Anchor a theory** that supports the study
3. **Guide** in developing relevant research objectives/questions
4. Help **justify** assumptions/hypothesis
5. Aid in choosing **appropriate methodology**
6. Help in gathering and interpreting data
7. Guide in identifying possible threat to validity



Differences of Theoretical and Conceptual Framework

	Theoretical Framework	Conceptual Framework
Content of Discussion	Focus on specific theory used already in the field	Focus on concepts related to your study
Range	Broad in sense Applicable in different studies	Narrower/focused Directly related to a specific study
Theories Presented	Presented one or more theory at a time and specifically related to the current study	May synthesize one or more theories or concept related to the current study
Time Frame	Focus on specific theory used already in the field	A working concept during the time of study

CONCEPT MAP

- visual representation of information that helps **show the relationship between ideas**
- begin with the main topic and then branch out into sub-topics, reflecting the connection of all the elements in the study
- can also provide an **organize new ideas**
- arranging ideas in a hierarchy

INPUT-PROCESS-OUTPUT MODEL

- a conceptual paradigm
- indicates the **inputs, required process**, and the **output**
- also referred to as a **functional model**
- used in action research where an **intervention or solution is necessary** to solve the identified problem

INPUT

- usually the independent variable

PROCESS

- intervention or solution consists of the instruments and analyses used to acquire the result

OUTPUT

- findings or outcome of the intervention being made to solve

INDEPENDENT VARIABLE-DEPENDENT VARIABLE MODEL (IDV)

- shows the independent variable and dependent variables of the study

RESEARCH HYPOTHESIS

HYPOTHESIS

- a **tentative statement** about the relationship between two or more variables
- **specific and testable prediction** on what you expect to happen
- **educated guess** of what possible the result would be of your research, which should be supported on existing theories and knowledge
- can **support or refute** through scientific research method
- **central concept** of any investigation that **converts research questions into predictions** and incorporates elements

Types of Research Hypothesis

1. SIMPLE HYPOTHESIS

- prediction of the relationship **between two variables**, the independent and dependent variables

2. COMPLEX HYPOTHESIS

- reflects the relationship between the independent variable and the dependent variable
- expresses connection **between two or more** independent variables and two or more variables

3. EMPIRICAL HYPOTHESIS

- "working hypothesis"
- presumed to explain certain facts and relationships of phenomena
- comes to life when a theory is being put to test, using **observation and experiment**
- can be changed or replace anytime
- going through some **trial and error**

4. LOGICAL HYPOTHESIS

- expresses explanation with **limited evidence** that can be verified logically
- it reflects a relationship of the variables which are anchored **based on logical phenomena**

5. STATISTICAL HYPOTHESIS

- analysis of a **portion of a population**
- can be verified statistically
- variables can be transformed into quantifiable sub-variable to assess it statistically

6. NULL HYPOTHESIS

- **H₀**
- exists when you consider that there is **no relationship** between the independent and dependent variables or that there is an **insufficient amount of information** to claim a scientific hypothesis

7. ALTERNATIVE HYPOTHESIS

- **H_a**
- expressed to be tested in order to **generate the desired output** when the empirical or working hypothesis is not accepted
- **disprove a null hypothesis**

PRACTICAL RESEARCH 2



DIRECTIONAL HYPOTHESIS

- related the relationship between the variables
- can also predict its nature
- illustrates the **direct association** of the impact of the independent variable with the dependent variable
- should be **clear and justified** according to the findings of the study
- investigated through a **one-tailed test**

NON-DIRECTIONAL HYPOTHESIS

- used when there is no principle involved
- the direction of the effect is **not specifically** determined
- reflects the association of the independent variable to the dependent variable without predicting the exact nature of the direction of the relationship
- investigated through **two-tailed test**

Guidelines on Formulating Hypothesis

1. Spend more time researching about the topic you are interested in. Focus on information and previous studies related to your topic.
2. Your independent variable and dependent variables must be included in your hypothesis.
3. The relationship of your variables must be reflected in your hypothesis.
4. Your hypothesis should be simple and specific as possible.
5. Should be concise and comprises clear and simple knowledge.
6. Should be testable without violating ethical standards.
7. Must be falsible. There are no absolute answers to research questions, but there is a possibility of validating the hypothesis to be true beyond a reasonable doubt.